Optimization Method for Urban Traffic Signal Control Based on Digital Twin Technology

Han Wang
School of Computer Science, Shenyang Aerospace University
Shenyang, China

Yunchong Guan*
School of Computer Science, Shenyang Aerospace University
Shenyang, China
y.c.guan@foxmail.com

Liang Zhao
School of Computer Science, Shenyang Aerospace University
Shenyang, China

Junling Shi
School of Computer Science, Shenyang Aerospace University
Shenyang, China

Shuo Li
School of Computer Science, Shenyang Aerospace University
Shenyang, China

Weizhi So
School of Computer Science, Shenyang Aerospace University
Shenyang, China

Jiaxuan Ma
School of Computer Science, Shenyang Aerospace University
Shenyang, China

Qi Song
School of Computer Science, Shenyang Aerospace University
Shenyang, China

Yue Zhang
School of Computer Science, Shenyang Aerospace University
Shenyang, China

Xiaobo Liu
School of Computer Science, Shenyang Aerospace University
Shenyang, China

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***Abstract*—Digital twin technology is promising, and has great advantage for traffic management to study, predict, simulate and experiment, from driving physical innovation to virtual testing. The goal remains the same: to improve traffic safety, regulate traffic flow, and ease urban congestion. At present, a mature digital twin transportation model has not been established at the social level. In this paper, a method is proposed to construct a digital twin traffic model by extracting environmental features and physical entities such as digital roads, traffic lights, and vehicles, and combining the optimization algorithm to take fuel consumption as the optimization goal, so as to provide ideas for improving the efficiency of urban road traffic and the quality of traffic environment.**

Based on the digital twin traffic map, combined with the genetic algorithm, this paper sets the objective function to fuel consumption, sets different iteration times, and performs multiple iterative optimizations on the four-phase duration of each intersection. After the optimization is completed, modify the phase duration of the original traffic light and run the simulation experiment to output the traffic information table file, according to which the user can select the optimal duration that matches the real environment. Finally, on the basis of genetic algorithm optimization, the green wave control is used to adjust the phase difference between the signal lights of the main road to further improve the optimization quality.

Keywords—environmental features, digital twin, transportation model, fuel consumption, green wave control

I. Introduction

The growth of urban population and the increase in per capital car ownership have posed significant challenges for urban traffic management. Traffic congestion brings great inconvenience to people's travel in real life, and direct testing of optimization algorithms in physical space can lead to a lot of waste. The article named Construction ideas and exploration of digital twin transportation system based on vehicle integration points out that digital twin technology is an effective method for traffic governance, but has not systematically proposed a method for constructing a digital twin model[1].A study has proposed Vehicle Self Organizing Network (VANET), which not only enables vehicles to communicate with nearby vehicles, but also exchanges data with nearby fixed devices through VANET[2]. This technology can obtain actual vehicle speed, location, gas consumption, and other information. The essence of urban transportation optimization methods based on digital twin technology is to combine digital twin technology with optimization algorithms, which are at the core of the digital twin technology framework. Therefore, the most obvious advantage of this combination method is to reduce cost expenditures and provide ideas for other industry research at present.To address this issue, this paper extracts key information elements from physical transportation, digitizes them, and uses software to construct a digital transportation model, which is then combined with optimization algorithms to reduce the cost of verification and improve efficiency[3].

There are several reasons why digital twins can be used for traffic management: Digital twins can virtually model cities, road networks, vehicles, etc. to simulate traffic flow.

Digital twins provide insight into traffic flow at different times and locations, helping to make traffic planning and management more accurate.

Digital twins can use historical and real-time data to predict traffic congestion. Traffic managers can use these predictions to take action to avoid or alleviate traffic congestion and improve traffic efficiency.

The digital twin can quickly establish the distribution and change of urban traffic flow through real-time collected data, providing the best reference for signal light regulation. In the simulated environment of the digital twin, traffic managers can

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test and optimize traffic signal schemes in real time and quickly implement them in real production environments to help solve traffic congestion problems.

The digital twin can simulate different traffic flow conditions and vehicle driving paths, and optimize road network design and urban planning schemes. Through the digital twin, different road network designs and traffic planning schemes can be quickly evaluated, and the optimal solution can be selected to improve traffic efficiency.

The paper presents a systematic method for constructing the model, providing ideas for other scholars to explore digital twin transportation models. This paper will be developed in four steps: using counting devices for road feature extraction, vehicle modeling, signal light configuration, and simulation interaction. Based on digital twin technology, a simulation model is established for traffic simulation, providing ideas for optimizing algorithm verification methods.

II. Road Characteristics Analysis

A. Scene Overview

To establish a digital twin model corresponding to physical transportation, it is necessary to extract key features from the environment, including road length, number of lanes, intersection length, vehicle driving direction (straight, left turn, right turn), and traffic signal light locations. This information is then imported into simulation software to initialize the twin model[4]. Using traffic flow counting devices, the road segments are surveyed on-site, and various factors such as vehicle speed, traffic volume, and start and end positions (origin and destination) for each vehicle during different time periods are set. At this point, the digital twin model is initialized.

B. Digital Twin Platform Interaction Analysis

The Digital twin platform interaction refers to the collaborative interaction between the twin layers[5].

The first layer involves analyzing task planning in physical space. By synchronizing physical tasks and setting digital tasks, the required variables for the twin model are identified. Based on the digital twin system, a three-dimensional model environment file that includes traffic signals, road length, number of lanes, speed limits, etc., is created to complete the task. The model is imported into the twin space, where it maps the model state of the physical space and changes synchronously with the variations in physical entity characteristics. As shown in Fig. 1.

The second layer involves using optimization algorithms to find appropriate signal light durations for the environment. By setting the objective function of increasing signal light utilization, the optimization algorithm aims to find the optimal solution of the objective function and feedbacks the optimized control duration to the twin body through I/O transmission.

Following the second layer, the third layer is the execution layer. After calculating the results from the previous step, the data is transferred to the twin model driver program to control the twin body to execute the corresponding operation. The twin model execution process is mutually mapped with the physical environment, and the feasibility of the optimization algorithm can be verified through the twin model. The rationality of the operation is evaluated manually, such as whether it adapts to local road environments and residents' living habits. After reasonable adjustments and twin scene changes, debugging is performed. When the intelligent agent fully adapts to the optimization algorithm, it indicates that the twin body under this task has completed the final task, and the optimization algorithm can be applied to the physical space.

C. Data Interaction

Interaction and collaboration refer to the real-time dynamic mapping of the physical entity's state in a virtual body, verifying control effectiveness through simulation in the virtual space, and generating insight feedback to the physical entity, forming a closed-loop for digital twins[6]. The interaction modes during signal light optimization include virtual-virtual interaction and physical-virtual interaction.

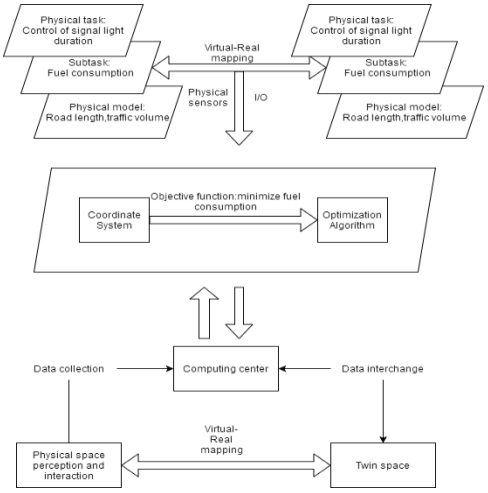

Fig.1. Joint schematic diagram of digital twins and optimization algorithms

(1) Virtual-virtual model: This interaction involves mutual communication between different virtual models, allowing for coordination among signal light controls in the area and achieving regional autonomy while optimizing individual signal lights. In this paper, virtual-virtual interaction mainly refers to the coordinated work of signal lights within an area controlled by the control algorithm, with communication files constantly collecting information from each signal light and transmitting it to the optimization algorithm. The algorithm dynamically changes the duration of some signal lights within the region to dynamically match the road communication of other areas.

(2) Physical-virtual model: This interaction involves synchronous changes between virtual models and physical entities, enabling physical entities to dynamically adjust based on the commands of the virtual model. By evaluating the performance of the optimization network through the physical environment, optimization algorithms dynamically adjust the network to better adapt to the physical environment. In this paper, physical-virtual interaction refers to sensors dynamically collecting data and transmitting it to the digital twin model to complete initialization; manual evaluation of the behavior of the twin model by the optimization algorithm is made promptly. During this process, the physical space and virtual space operate in a collaborative manner.

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III. Digital Twin Modeling and Simulation Methods

A. Digital Twin Model Data Acquisition

To establish a digital twin three-dimensional model based on extensive feature data, data is obtained through physical sensing devices. Vehicle flow meters are set up to record actual intersection traffic volume, vehicle speed, and other information during the green light period and used as input for the digital twin model's initialization. To continuously obtain traffic volume data, vehicles pass through the intersection from the north, south, east, and west directions. The vehicle flow meter continuously collects the number of left-turn and through vehicles at each direction during the green light duration and inputs them into their corresponding signal light optimization algorithms. The number of right-turn vehicles does not need to wait for the signal light and have to give way to the through vehicles, so the number of right-turn vehicles does not need to be calculated. The counter's counting duration changes with the signal light duration, continually updating the signal light duration re-determined by the optimization algorithm and obtaining data as input for the digital twin model.

To make the road situation more intuitive and the twin model more image-like, this paper uses the open-source, multimodal traffic simulation software SUMO for city road state modeling[7]. OpenStreetMap is selected from the open-source wiki map to download the required city area map file and import it into SUMO to replace the default map, including actual information such as road length. Since this research focuses on the optimization algorithm under traffic volume, a map with traffic flow needs to be obtained. Through field investigation, the vehicle speed information is set in SUMO for each vehicle and the signal light position, start and end positions of each vehicle are configured, converting physical data into digital data for use in the twin model.

B. Simulation Plan

First, download the SUMO software and configure the software environment variable to the SUMO installation path. It should be noted that the versions of SUMO, Veins, and Omnet++ must correspond to each other. Next, download the area terrain file in .osm (geographic coordinates) format through OpenStreetMap. Use the net conversion file in Sumo to input the map file elements, parse out the pure text format digital road network file, and output it. The digital road network file (Map.net.xml) includes node and edge files, which represent the road network file drawn using points and edges[8]. The node file includes road intersections, which are located by coordinates x and y. Each edge in the edge file is a one-way lane corresponding to a one-way lane, with edge information including the start and end nodes of the edge, intersection signal light position, name, number of lanes, etc.

Furthermore, configure vehicle behavior through the configuration file, generate the vehicle's driving path and speed, integrate roads and paths to form the vehicle's driving behavior, and create the vehicle flow file (Trips.xml), which includes traffic volume and start and end positions. Each vehicle has a separate speed factor, which, when multiplied by the speed limit, determines the required travel speed. By reasonably configuring SpeedFactor and SpeedDev attributes to individually set the speed factors of each vehicle as a fixed value, the final product meets the data range obtained from field investigations. Each vehicle is configured, and corresponding road flows are generated when multiple vehicles exist on the same road.

Furthermore, after configuring the traffic volume, the traffic needs to move forward according to the signal light instructions; otherwise, traffic accidents such as rear-end collisions may occur. Therefore, the Trips.xml file generated above needs to be configured for signal lights. The detailed signal light configuration process can be set through the netedit file in SUMO, considering the following traffic scenario. The intersections are marked with different digital IDs, and the signal lights in each direction cooperate with each other.

The signal light status corresponds one-to-one with the signal light during that period, and the same signal light type (type) is set in each direction. The ID coding sequence strictly follows from the road in the 12 o'clock direction to the left turn path, clockwise rotation through the middle path. The main traffic signal lights used in China have three colors: red, yellow, and green, corresponding to r, y, and G in English, respectively. There are three ways for vehicles to turn: straight, right, and left, and there are three directions of turning on the same lane. Because of the left turn's priority, to give priority to through traffic, the left turn green light is set to g, and the through green light is set to G. Therefore, the signal light configuration is formed by combining rygG with each other. Hence, G has the highest priority. The signal lights in the same intersection cooperate with each other strictly according to the encoding sequence. As shown in Fig. 2, the signal lights in this scenario correspond to rrrrGGGggrrrrGGGgg.

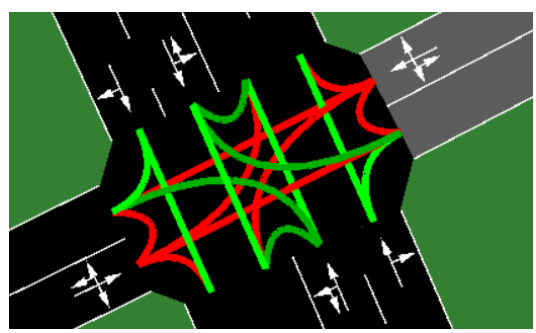

Fig.2. Digital traffic intersection

Furthermore, after completing the above steps, a configuration file (Map.sumo.cfg) is written for the map file, which saves the log files and software configurations of the road network running program. The result can be viewed by running the sumo-gui configuration file to open the SUMO visualization interface. The digital traffic running is shown in Fig. 3.

Finally, to run the generated digital traffic through Omnet++, a poly.xml terrain file needs to be generated based on the map file and terrain file and loaded into Veins. Copy the road network file, traffic flow file, terrain file, and configuration file into Veins, and call the digital traffic configuration file in the initialization function (.ini) of Veins to run under the omnet++ platform. The signal light duration changes and traffic flow changes controlled by the optimization algorithm can be observed using the Traci interface.

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The Traci interface needs to be started directly in SUMO and supports multiple languages such as Python and Java. It can extract various indicators such as nodes, roads, detectors, signal lights, and vehicles in real-time during the simulation process. When combined with the optimization algorithm, the algorithm can directly control it and thus control the traffic situation in the simulation model and change the traffic situation.

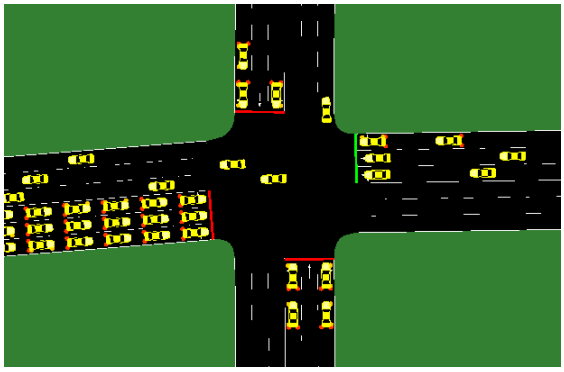

Fig.3. Vehicle fuel consumption index

C. *Optimization Process*

The optimization process in this study adopts the divide-and-conquer approach[9], first optimizing the individual signal lights separately and then selecting a group of signal lights in the area for green wave control optimization to achieve multi-objective optimization purposes.

Use genetic algorithms to optimize individual semaphores. Genetic algorithms are computational models that simulate the selection, mutation, and genetic mechanisms of biological evolution in Darwinian evolution. It has the advantage of selecting values closer to the optimal solution each time for crossover and mutation to obtain more qualified offspring, which helps solve the optimal problem[10].

This optimization process chooses single-objective optimization and sets the optimization target as vehicle fuel consumption[11]. When the fuel consumption is lower, the waiting time at the intersection is shorter during the same driving distance. Traditional optimization targets are expressed in the form of mathematical equations. The concept of "adaptive traffic signal control algorithm based on green computing" has been proposed at the social level, also known as sustainable computing, aiming to greatly reduce energy consumption in the process of using computer hardware and software[12]. This paper will use real-time obtained vehicle fuel consumption index in the twin model as the standard for selecting optimization target elements, making algorithms to achieve cleaner and more efficient optimization results and making the algorithm more flexible in interacting with vehicles. By obtaining the real-time value of fuel consumption, the subsequent crossover and mutation genetic processes select the duration with smaller fuel consumption and output the optimal signal light duration. Fig. 3 shows real-time data of fuel consumption index under the optimization algorithm. This paper sets the fitness function as the total duration of the four phases of the signal lights. The difference between its value and the corresponding total duration of the original signal lights' four phases is close to zero, representing a higher fitness of the solution element, i.e., a better solution. During interaction with Sumo, the program can call real-time information of each vehicle in the twin model, such as the starting and ending positions, travel time, and fuel consumption, and output them as a table file for comparison with data before the optimization stage to observe their trends.

Signal light group optimization: A research has proposed an optimization method based on ant colony optimization algorithm, which reduces traffic congestion by selecting the best path for vehicles during peak hours[13]. This article changes the duration of the signal lights in the area without changing the vehicle route, so that each green light cooperates with each other. In this study, green wave control is used to optimize signal light groups in the area. The advantage of green wave control is its simplicity in modifying signal light durations and allowing vehicles to pass continuously through a series of green lights in the area during non-traffic congestion periods without causing an increase in congestion index due to vehicle delays, which has ability to achieve traffic signal optimization[14].

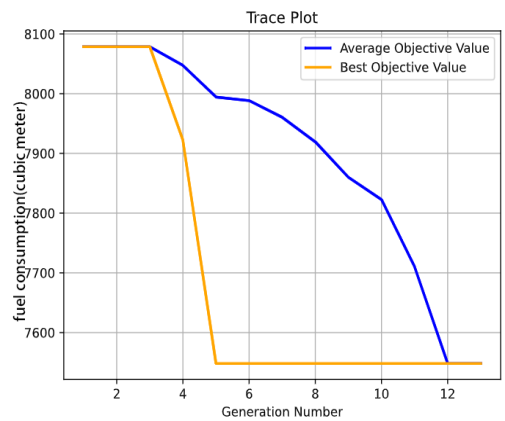

Fig.4. Vehicle fuel consumption index

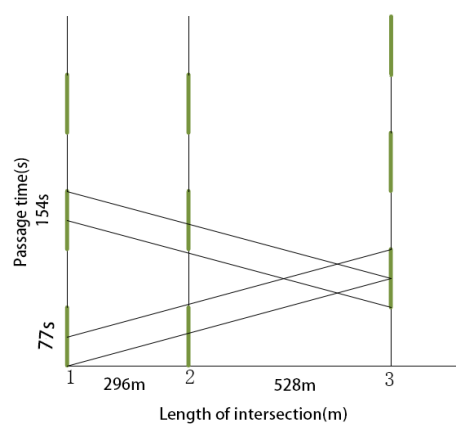

Fig.5. Graphic method bidirectional green wave control

The purpose of multi-objective optimization is achieved by using green wave control. There are mainly two types of green wave control methods: graphic method and numerical method. For more intuitive and convenient observation and operation, this paper chooses the graphic method for bidirectional green wave control of signal lights. Bidirectional green wave control refers to changing the signal light phase on the north-south/south-north and east-west/west-east bidirectional lanes. Phase difference: the time difference between the start or middle

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point of the green or red light of each signal light and the start or middle point of the standard signal green or red light[15].

Using the signal light duration obtained from the above optimization algorithm as the Y-axis and the distance between each intersection as the X-axis, a rectangular coordinate system is drawn. Taking the first signal light as the reference, it is determined whether the subsequent controlled signal lights belong to direct control or interactive control with the reference signal light. The phase data of each signal light is then selected through the coordinate system. The obtained phase data and optimized duration are combined and configured on each signal light to form the optimized duration of the complete twin model. Fig. 4 shows the schematic diagram of graphic method green wave control, taking the east-west/west-east direction as the bidirectional control object, and intuitively obtaining the start or end time difference between the signal light and the reference signal light. In the above optimization process, interaction with the twin model is required at each step to observe the traffic model by controlling the twin model through data input. After the optimization process is completed, the table file can be observed to determine whether the optimization process is suitable. If it is determined through human observation that this optimization method is applicable to the twin model, it can be used in physical space.

IV. Conclusion

This paper aims to provide a method for building a systematic digital twin model and reduce the cost of physical testing while improving inspection efficiency with its ease of operation.

In practice, the modeling process of a digital twin traffic model is relatively easy to achieve, and when combined with optimization algorithms to control digital traffic, the twin model exhibits advantages such as ease of operation and high efficiency compared to the original model. By continuously training the twin body through the combination of a digital twin model and optimization algorithm to obtain the optimal solution, after completing the training, the physical entity can be controlled to operate.

In short, the role of digital twins in traffic management is very important, which can help transportation planners better understand urban and road traffic data, better formulate traffic planning, optimize traffic signals and road network design, improve traffic efficiency, improve traffic safety, and provide a higher level of urban management services for cities.This approach is not only applicable to twin traffic models but also can improve efficiency and reduce costs in other fields of inspection by constructing a twin body.

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